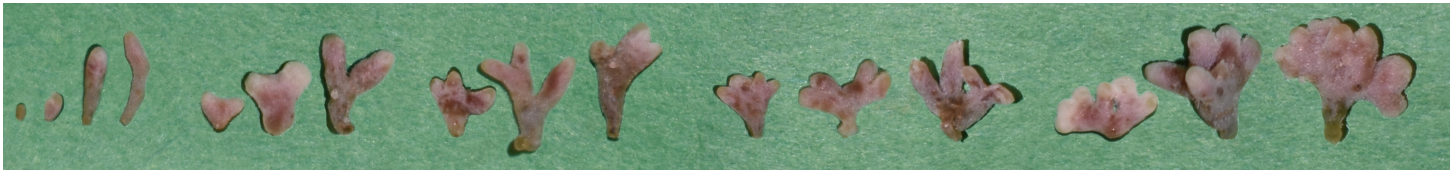
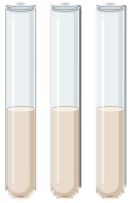
A) Branching Variation Across 3 alfalfa varieties



B) Experimental Overview

1. Inoculum prepared
2. Single-Isolate Inoculations
3. Greenhouse experiment

117 *S. meliloti strains*



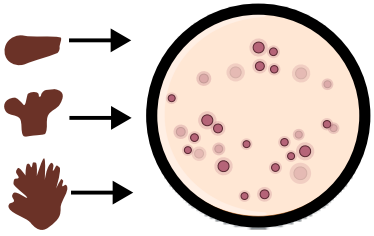
3 alfalfa varieties

Oneida (O) SW402(S) Vernal (V)

- Four replicates
- Random block design
- No N fertilizer added
- Grown for 2 months

C) Rhizobial Benfits

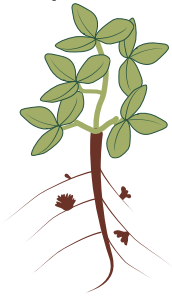
8 Isolates subset - traits measued:

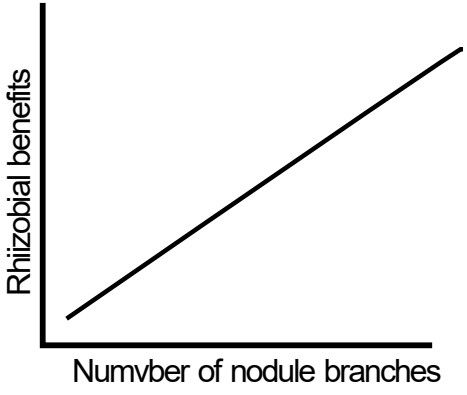
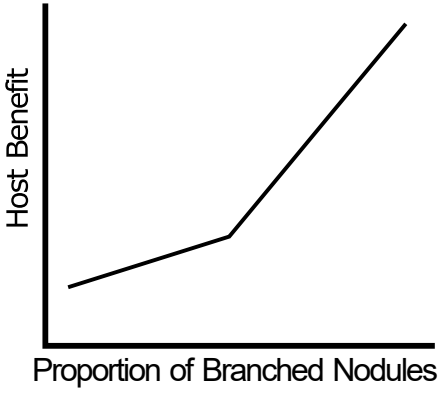


- Nodule Surface Area
- # of Nodule branches
- # CFUs released per cursched nodule

D) Plant Benefits

117 plants - traits measured:



- Plant biomass
- Nodule number
- Number of branched nodules

E) Hypothesized Relationships